```python
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder

data = pd.read_csv("breast_cancer.csv")
```

```
PS C:\my files\Ai & ml internship files>  python breast.py
           id diagnosis  radius_mean  texture_mean  ...  concavity_worst  concavepoints_worst  symmetry_worst  fractal_dimension_worst
0      842302         M        17.99         10.38  ...           0.7119               0.2654          0.4601                  0.11890
1      842517         M        20.57         17.77  ...           0.2416               0.1860          0.2750                  0.08902
2    84300903         M        19.69         21.25  ...           0.4504               0.2430          0.3613                  0.08758
3    84348301         M        11.42         20.38  ...           0.6869               0.2575          0.6638                  0.17300
4    84358402         M        20.29         14.34  ...           0.4000               0.1625          0.2364                  0.07678
..        ...       ...          ...           ...  ...              ...                  ...             ...                      ...
564    926424         M        21.56         22.39  ...           0.4107               0.2216          0.2060                  0.07115
565    926682         M        20.13         28.25  ...           0.3215               0.1628          0.2572                  0.06637
566    926954         M        16.60         28.08  ...           0.3403               0.1418          0.2218                  0.07820
567    927241         M        20.60         29.33  ...           0.9387               0.2650          0.4087                  0.12400
568     92751         B         7.76         24.54  ...           0.0000               0.0000          0.2871                  0.07039

[569 rows x 32 columns]
PS C:\my files\Ai & ml internship files>
```

```python
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder

data = pd.read_csv("breast_cancer.csv")
label_encoder=LabelEncoder()
data["diagnosis_labelled"] = label_encoder.fit_transform(data["diagnosis"])
print(data)
```

```
..       ...       ...      ...       ...  ...        ...         ...         ...          ...
564   926424         M    21.56     22.39  ...     0.4107      0.2216      0.2060      0.07115
565   926682         M    20.13     28.25  ...     0.3215      0.1628      0.2572      0.06637
566   926954         M    16.60     28.08  ...     0.3403      0.1418      0.2218      0.07820
567   927241         M    20.60     29.33  ...     0.9387      0.2650      0.4087      0.12400
568    92751         B     7.76     24.54  ...     0.0000      0.0000      0.2871      0.07039

[569 rows x 32 columns]
PS C:\my files\Ai & ml internship files> python breast.py
           id diagnosis  radius_mean  texture_mean  ...  concavepoints_worst  symmetry_worst  fractal_dimension_worst  diagnosis_labelled
0      842302         M        17.99         10.38  ...               0.2654          0.4601                  0.11890                   1
1      842517         M        20.57         17.77  ...               0.1860          0.2750                  0.08902                   1
2    84300903         M        19.69         21.25  ...               0.2430          0.3613                  0.08758                   1
3    84348301         M        11.42         20.38  ...               0.2575          0.6638                  0.17300                   1
4    84358402         M        20.29         14.34  ...               0.1625          0.2364                  0.07678                   1
..        ...       ...          ...           ...  ...                  ...             ...                      ...                 ...
564    926424         M        21.56         22.39  ...               0.2216          0.2060                  0.07115                   1
565    926682         M        20.13         28.25  ...               0.1628          0.2572                  0.06637                   1
566    926954         M        16.60         28.08  ...               0.1418          0.2218                  0.07820                   1
567    927241         M        20.60         29.33  ...               0.2650          0.4087                  0.12400                   1
568     92751         B         7.76         24.54  ...               0.0000          0.2871                  0.07039                   0

[569 rows x 33 columns]
PS C:\my files\Ai & ml internship files>
```

breast_cancer.csv | breast.py | SUPPL 6

breast.py > ...

```python

data = pd.read_csv("breast_cancer.csv")
label_encoder=LabelEncoder()
data["diagnosis_labelled"] = label_encoder.fit_transform(data["diagnosis"])

X = data [["radius_mean","texture_mean","perimeter_mean","area_mean","smoothness_mean","compactness_mean","concavity_mean","concavepo
Y = data[["diagnosis_labelled"]]
print(X)
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
..      ...    ...     ...     ...  ...      ...      ...       ...   ...
564  926424      M   21.56   22.39  ...   0.2216   0.2060   0.07115     1
565  926682      M   20.13   28.25  ...   0.1628   0.2572   0.06637     1
566  926954      M   16.60   28.08  ...   0.1418   0.2218   0.07820     1
567  927241      M   20.60   29.33  ...   0.2650   0.4087   0.12400     1
568   92751      B    7.76   24.54  ...   0.0000   0.2871   0.07039     0

[569 rows x 33 columns]
PS C:\my files\Ai & ml internship files> python breast.py
     radius_mean  texture_mean  perimeter_mean  area_mean  ...  concavity_worst  concavepoints_worst  symmetry_worst  fractal_dimension_worst
0          17.99         10.38          122.80     1001.0  ...           0.7119               0.2654          0.4601                  0.11890
1          20.57         17.77          132.90     1326.0  ...           0.2416               0.1860          0.2750                  0.08902
2          19.69         21.25          130.00     1203.0  ...           0.4504               0.2430          0.3613                  0.08758
3          11.42         20.38           77.58      386.1  ...           0.6869               0.2575          0.6638                  0.17300
4          20.29         14.34          135.10     1297.0  ...           0.4000               0.1625          0.2364                  0.07678
..           ...           ...             ...        ...  ...              ...                  ...             ...                      ...
564        21.56         22.39          142.00     1479.0  ...           0.4107               0.2216          0.2060                  0.07115
565        20.13         28.25          131.20     1261.0  ...           0.3215               0.1628          0.2572                  0.06637
566        16.60         28.08          108.30      858.1  ...           0.3403               0.1418          0.2218                  0.07820
567        20.60         29.33          140.10     1265.0  ...           0.9387               0.2650          0.4087                  0.12400
568         7.76         24.54           47.92      181.0  ...           0.0000               0.0000          0.2871                  0.07039

[569 rows x 30 columns]
PS C:\my files\Ai & ml internship files>
```

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```python
data = pd.read_csv("breast_cancer.csv")
label_encoder=LabelEncoder()
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X = data [["radius_mean","texture_mean","perimeter_mean","area_mean","smoothness_mean","compactness_mean","concavity_mean","concavepo
Y = data[["diagnosis_labelled"]]
print(Y)
```

```
..        ...      ...      ...     ...  ...        ...         ...        ...          ...
564     21.56    22.39   142.00  1479.0  ...     0.4107      0.2216     0.2060      0.07115
565     20.13    28.25   131.20  1261.0  ...     0.3215      0.1628     0.2572      0.06637
566     16.60    28.08   108.30   858.1  ...     0.3403      0.1418     0.2218      0.07820
567     20.60    29.33   140.10  1265.0  ...     0.9387      0.2650     0.4087      0.12400
568      7.76    24.54    47.92   181.0  ...     0.0000      0.0000     0.2871      0.07039

[569 rows x 30 columns]
PS C:\my files\Ai & ml internship files>  python breast.py
     diagnosis_labelled
0                     1
1                     1
2                     1
3                     1
4                     1
..                  ...
564                   1
565                   1
566                   1
567                   1
568                   0

[569 rows x 1 columns]
PS C:\my files\Ai & ml internship files>
```

File Edit Selection View Go Run Terminal Help

Ai & ml internship files

EXPLORER

AI & ML INTERNSHIP FILES

- breast_cancer.csv
- breast.py
- knn_data.csv
- knn.py
- load_extension.csv
- machine_fail.py
- machine_failure.csv
- pgm.py
- power_data.csv
- Power.py
- SUPPL 6

breast_cancer.csv | breast.py | SUPPL 6

breast.py > ...

```python
label_encoder=LabelEncoder()
data["diagnosis_labelled"] = label_encoder.fit_transform(data["diagnosis"])

X = data [["radius_mean","texture_mean","perimeter_mean","area_mean","smoothness_mean","compactness_mean","concavity_mean","concavepo
Y = data[["diagnosis_labelled"]]

model= LogisticRegression()
model.fit(X,Y)
print("coeff:",model.coef_)
print("intercerpt:",model.intercept_)
prediction= model.predict([[17.99,10.38,122.8,1001,0.1184,0.2776,0.3001,0.1471,0.2419,0.07871,1.095,0.9053,8.589,153.4,0.006399,0.049
print("predicted cancer type:",prediction)
if prediction == 0:
    print("BENGIN")
else:
    print("MALIGANT")
```

PROBLEMS 6 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
coeff: [[-0.82602533 -0.39068411 -0.36582212  0.01686076  0.02930868  0.1406838
   0.19725844  0.08284938  0.04070186  0.00802399 -0.03490221 -0.31465127
  -0.09847498  0.10514417  0.00268178  0.03053159  0.04240827  0.01084177
   0.00985935  0.00281148 -0.87558153  0.45080564  0.294833    0.01244202
   0.05352652  0.44098701  0.54884048  0.15995218  0.13017132  0.04200612]]
intercerpt: [-0.15245352]
C:\Users\Parthajith KS\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
predicted cancer type: [1]
MALIGANT
```

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

breast.py

Describe what to build